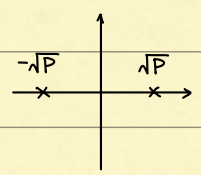
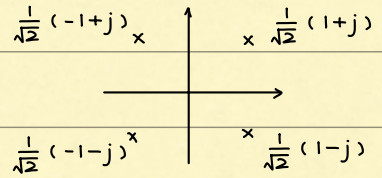
Constellation 星座图



BPSK



QPSK

$$\text{Energy } \mathcal{E} = \frac{1}{M}\sum_{m=1}^{M} E(|s(n)|^2)$$

AWGN Channel $y = x + w$

$w \sim CN(0, N_0)$

$w = Re(w) + jIm(w)$ $\quad Re(w), Im(w) \sim N(0, \frac{N_0}{2})$

BPSK $x = \pm\sqrt{P}$ $\quad P_e = Q(\sqrt{P/\sigma^2})$

Noise $\frac{N_0}{2}$, $SNR = \frac{P}{N_0}$, $\quad P_e = Q(\sqrt{2SNR}) \propto e^{-SNR}$

$\hat{x}_{ML}$ 取最近

Rayleigh Fading Channel

$y = hx + w$, $\quad h \sim N(0,1)$

若 h 已知, Coherence Direction $r = \frac{h^*}{|h|}y$

$\hat{x}_{ML}$ 为取 r 离哪个码最近

$P_e = Q(\sqrt{|h|^2 P/N}) = Q(\sqrt{2|h|^2 SNR})$

$|h|^2 = (Re\,h)^2 + (Im\,h)^2 \sim \chi^2(2) = Exp(1)$

e.g. QPSK, $y = 1-j$, $\quad h = 3+4j$

$r = \frac{h^* y}{|h|} = \frac{1}{5}(3-4j)(1-j)$

$= \frac{1}{5}(-1-7j) \Rightarrow \hat{x}_{ML} = \frac{1}{\sqrt{2}}(-1-j)$

Diversity 分集

为解决 Multipath 的 selective fading 带来的 ISI

MA (Multiple Access) 多址技术

TDMA 分时多址技术

1. Repetition Code

$y[n] = h[n]x + w[n]$, $\quad N$: codelength

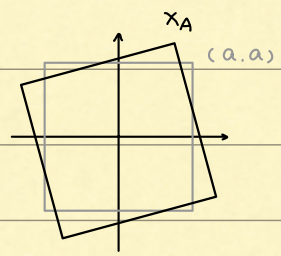
Match Filter / MRC $\quad \tilde{y} = \frac{1}{\|h\|} h^H y$

$$= \|h\| x + \tilde{w}$$

$P_e = Q(\sqrt{\|h\|^2 P/N}) \qquad \|h\|^2 \sim \chi^2(2N)$

$$\approx \binom{2N-1}{N} \frac{1}{(4\,SNR)^N}$$

2. Rotation Code



$$y_{2\times1} = \underbrace{h_{2\times1} \odot x_{2\times1}}_{u_{2\times1}} + w_{2\times1}$$

$x_A = R\begin{bmatrix} a \\ a \end{bmatrix} \qquad x_B = R\begin{bmatrix} -a \\ a \end{bmatrix}$

$$Pr(x_A \to x_B) = Q\left(\frac{\|u_A - u_B\|/2}{\sqrt{N_0/2}}\right)$$

$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$$u_A - u_B = h \odot a \cdot \underbrace{\frac{1}{a}(x_A - x_B)}_{\text{Normalized } d = 2\begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}} = a\, h \odot d$$

$$= Q\left(\sqrt{\frac{SNR}{2}(h_1^2 d_1^2 + h_2^2 d_2^2)}\right)$$

$$\leq \frac{16}{|d_1 d_2|^2} SNR^{-2}, \qquad \delta_{AB} = |d_1 d_2|^2$$

Antenna Diversity / Spacing Diversity

measured SNR $\gamma \sim Exp(\frac{1}{\bar{\gamma}})$ in Rayleigh Fading

PDF $\frac{1}{\bar{\gamma}} e^{-\gamma/\bar{\gamma}}$

CDF $1 - e^{-\gamma/\bar{\gamma}}$

Outage Probability $\quad P_{out} = Pr(\gamma < \gamma_0) = 1 - e^{-\frac{\gamma_0}{\bar{\gamma}}}$

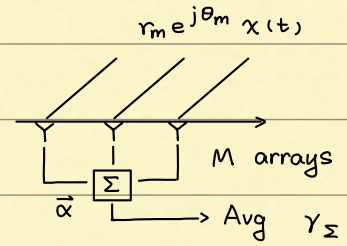
γ_m : 实测的 SNR

$\bar{\gamma}$: Fading 带来的平均 SNR $\quad \gamma_m \sim Exp(\frac{1}{\bar{\gamma}})$

γ_Σ : 用某种方法 SC/MRC/EGC 得的 RV

CDF $\quad P_{\gamma_\Sigma} = Pr(\gamma_\Sigma < \gamma)$

PDF $\quad p_{\gamma_\Sigma}(\gamma)$



$\bar{\gamma}_\Sigma$: 由 γ_Σ 的 PDF 得的均值

A_g : Average Gain $\quad \bar{\gamma}_\Sigma / \bar{\gamma}$

γ_0 : 中断阈值

SC $\quad \alpha = (0, \cdots, 1, \cdots, 0)$

MRC $\quad \alpha = h$

EGC $\quad \alpha = h / \|h\|$

P_{out} : $Pr(\gamma_\Sigma < \gamma_0)$, 即 CDF 代入 γ_0

1. Receiver Combining Technique

SC (Selection Combining) $\qquad$ Downlink 下行端

$\gamma_\Sigma = \max \gamma_m$

CDF $\quad P_{\gamma_\Sigma}(\gamma) = Pr(\gamma_\Sigma < \gamma) = \prod_{m=1}^{M} Pr(\gamma_m < \gamma)$

$= (1 - e^{-\gamma/\bar{\gamma}})^M$

$P_{out} = P_{\gamma_\Sigma}(\gamma_0) = (1 - e^{-\gamma_0/\bar{\gamma}})^M$

PDF $\quad p_{\gamma_\Sigma}(\gamma) = \frac{d}{d\gamma} P_{\gamma_\Sigma}(\gamma)$

$\bar{\gamma}_\Sigma = \int \gamma \, p_{\gamma_\Sigma}(\gamma) \, d\gamma = \bar{\gamma} \cdot \sum_{m=1}^{M} \frac{1}{m}$

$A_g = \sum_{m=1}^{M} \frac{1}{m}$

MRC (Maximal Ratio Combining)

Weight $\quad \alpha_m = r_m e^{-j\theta_m}$

$\gamma_\Sigma = \sum_{m=1}^{M} \gamma_m \sim \chi^2(2M)$

$A_g = M$

EGC (Equal Gain Combining)

Weight $\quad \alpha_m = e^{-j\theta_m}$

$A_g = [1 + (M-1)\frac{\pi}{4}]$

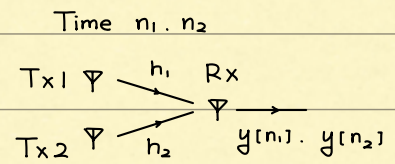
2. Trasmitter Diversity

Alamouti Scheme / OTSBC 不依赖 CSI

2个发射，1个接受，2个时间拍

在 n_1 时刻，

$$y[n_1] = h_1 x_1 + h_2 x_2$$



在 n_2 时刻，发送 $[-x_2^* \quad x_1]$

$$y[n_2] = h_1(-x_2^*) + h_2 x_1^*$$

$$= (h_2^* x_1 - h_1^* x_2)^*$$

$$\underbrace{\begin{bmatrix} y[1] \\ y^*[2] \end{bmatrix}}_{y_{2\times1}} = \underbrace{\begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix}}_{H_{2\times2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} w[1] \\ w^*[2] \end{bmatrix}$$

$$z_{2\times1} = H^H y = H^H H x + H^H w$$

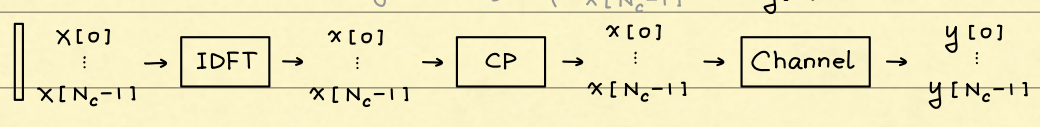
$$= (|h_1^2| + |h_2^2|)\, x + \tilde{w}$$

$$SNR = (|h_1^2| + |h_2^2|)\frac{E_s}{2N_0}$$

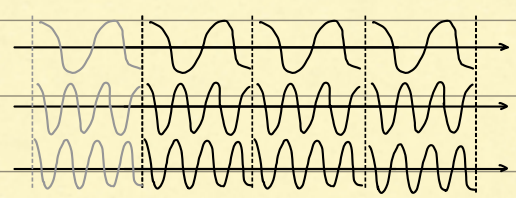
OFDM (Orthogonal Frequency Diversity Multiplexing)

Cyclic Prefix $\left\{ \begin{matrix} x[N_c-2] \\ x[N_c-1] \end{matrix} \right.$

$y[n] = h[n]x[n] + w[n]$



Overhead $\frac{L_c}{L_c+N_c}$



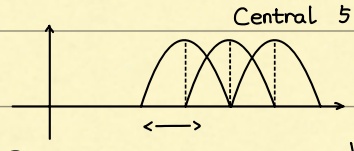
Symbol Period T_s

Guard interval $L_c T_s$ $\quad T_N = N_c T_s$

OFDM Period $T_{OFDM} = (N_c + L_c) T_s$

Subcarrier Number N_c

Central 5 GHz



Subcarrier Spacing $\Delta = \frac{W}{N_c} = \frac{1}{T_N}$

Total Bandwidth $W = \frac{1}{T_s}$

Rate $R_{min} = \log_2 M \cdot r_{code} \cdot N_c / T_{OFDM}$

No ISI $T_d \ll L_c T_s$

No Doppler $T_c \gg T_{OFDM}$